

Cube Roots — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (a) **10.** $10^3 = 1000$.

Q2 (a) **9.** $9^3 = 729$.

Q3 **4.** $4^3 = 64$.

Q4 (c) **8.** $8^3 = 512$ ($512 = 2^9$).

Q5 **True.** $3^3 = 27$.

Q6 (d) **A false, R true.** $\sqrt[3]{1728} = 12$, whose ones digit is 2 (not 6), so A is false; the reason is a correct definition.

Short answer

Q7 **13.** $2197 = 13 \times 13 \times 13$.

Q8 **6.** $216 = (2 \times 2 \times 2)(3 \times 3 \times 3)$, so $\sqrt[3]{216} = 2 \times 3 = 6$.

Q9 **18.** $5832 = 2^3 \times 3^6 = (2 \times 2 \times 2)(3 \times 3 \times 3)(3 \times 3 \times 3)$, so $\sqrt[3]{5832} = 2 \times 3 \times 3 = 18$.

Long answer

Q10 **12.** $1728 = (2 \times 2 \times 2)(2 \times 2 \times 2)(3 \times 3 \times 3)$, so $\sqrt[3]{1728} = 2 \times 2 \times 3 = 12$.

Q11 **24.** $13824 = 2^9 \times 3^3$, so $\sqrt[3]{13824} = 2^3 \times 3 = 8 \times 3 = 24$.

Q12 **Edge = 15 cm.** $\text{Edge} = \sqrt[3]{3375} = \sqrt[3]{[(3 \times 3 \times 3)(5 \times 5 \times 5)]} = 3 \times 5 = 15$.

Total: 21 marks.